Year	Dist_Name	RICE_AREA _1000_ha	RICE_PRODUC-TION_1000_tons	RICE YIELD Kg_per_ha
1966	Burdwan	436.48	597.28	1368.4
1967	Burdwan	459.82	626.65	1364.61
1968	Burdwan	473.8	687.22	1454.51
1969	Burdwan	476.1	629.26	1322.7
1970	Burdwan	463.2	617.88	1332.78
1971	Burdwan	500.8	616.16	1233.81
1972	Burdwan	496	709.42	1430.28
1973	Burdwan	526.2	664.17	1262.2
1974	Burdwan	535.4	806.35	1506.07

Table 1: Rice production data of some districts of the state of West Bengal

	Year	Dist_Name	RICE_AREA_1000_ha	RICE_PRODUCTION_1000_tons	RICE_YIELD_Kg_per_ha
0	1966	Burdwan	436.48	597.28	1368.40
1	1967	Burdwan	458.82	625.65	1363.61
2	1968	Burdwan	472.80	687.22	1453.51
3	1969	Burdwan	476.10	629.26	1321.70
4	1970	Burdwan	463.20	616.88	1331.78

Figure 1: Rice production

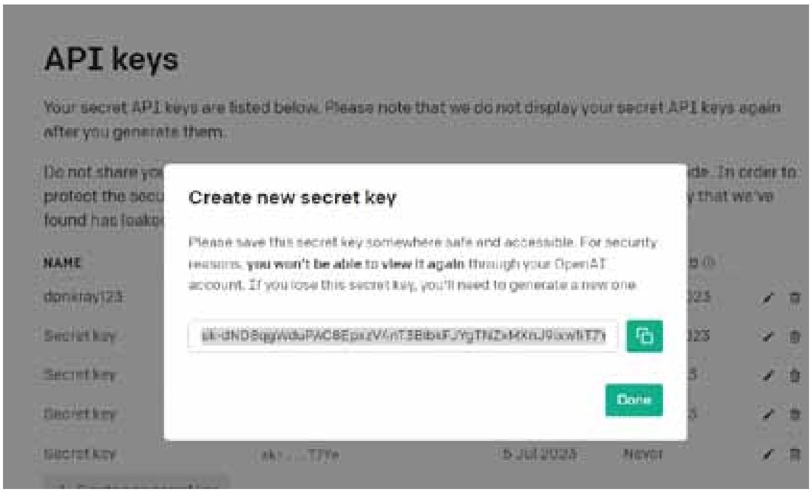

Figure 2: Generation of API key from ChatGPT

```
   Year Dist_Name  RICE_AREA_1000_ha  RICE_PRODUCTION_1000_tons  \
0  1966   Burdwan             436.48                     597.28
1  1967   Burdwan             458.82                     625.65
2  1968   Burdwan             472.80                     687.22
3  1969   Burdwan             476.10                     629.26
4  1970   Burdwan             463.20                     616.88

   RICE_YIELD_Kg_per_ha
0               1368.40
1               1363.61
2               1453.51
3               1321.70
4               1331.78
```

Figure 3: Output in tabular form

Once this command is executed, Python will request an input file. A CSV file named *CRISAT-WB_Rice.csv* has been uploaded in this instance. The file contains 364 record sets detailing rice production in various districts of West Bengal. Interested readers can download the file using the link provided at the end of the article.

Exploring data with PandasAI

After uploading the input CSV file, proceed to read the CSV file from memory:

```
import pandas as pd
import io
df = pd.read_csv(io.
StringIO(uploaded['ICRISAT-WB_Rice.csv'].
decode('utf-8')))

df.head()
```

The output is shown in Figure 1.

To call *pandasai()*, it is required to have a unique OpenAI API key. Obtain the key by creating an OpenAI account from *https://openai.com/blog/chatgpt*. From there it is possible to create an API key as shown here in Figure 2.

Once you have this key, you can call these two functions to start PandasAI:

```
oail = OpenAI(api_token="Insert your API-
KEY here")
pandas_ai = PandasAI(oali,
conversational=False)
```

For the 'True' value of the conversational option, PandasAI delivers a descriptive text record set, whereas it returns a tabular set of data sets for the 'False' value.

Working with Pandas DataFrame

The index of a record can be explored with the *pandas_ai()* call:

```
# Finding the index of a row using the value
of a column
result = pandas_ai(df, "What is the index
of 1982?")
print(result)
16
```